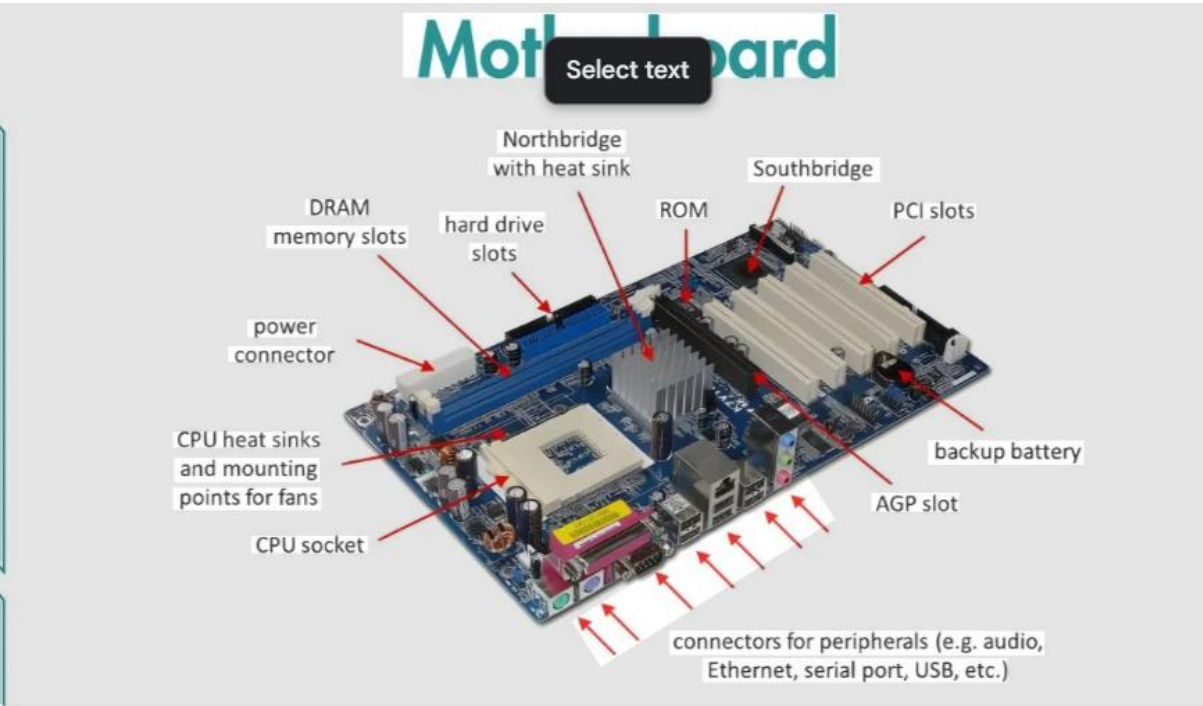


1. Labelled Diagram of an ATX Motherboard



2. Popular CPU Sockets & Matching Processors

Socket Type	Brand	Supported CPU Generations	Example CPU
AM4	AMD	Ryzen 1000 through 5000 series	AMD Ryzen 5 5600X
LGA1700	Intel	12th, 13th, and 14th Gen Core series	Intel Core i5-13600K
AM5	AMD	Modern Ryzen 7000, 8000, & 9000 series	AMD Ryzen 7 7800X3D

3. How the CPU and Chipset Work Together (Opening Spotify)

What They Do

- **CPU:** The overall brain of the computer that handles logic, runs software instructions, and processes commands.

- **Chipset:** The motherboard's traffic cop. It manages the flow of data between the CPU and all secondary parts like hard drives, USB ports, sound cards, and network connections.

What Happens When You Open Spotify

1. **The Click:** You double-click Spotify. Your mouse sends a physical signal through a **USB port**, which the **chipset** receives and hands off to the **CPU**.
2. **Loading the Files:** The **CPU** recognizes the command and requests Spotify's program files. The **chipset** fetches those files from your SSD or hard drive and loads them into your RAM.
3. **Playing the Music:** The **CPU** launches the app onto your screen and routes the music playback data back through the **chipset** to your speakers or headphones.

4. Real Motherboard Ports Identified

5 Main Ports Identified:

1. **USB 2.0 Ports (Black, far left):** Standard USB ports typically used for connecting basic peripherals like your mouse and keyboard.
2. **HDMI / Display Port (Bottom left):** Plugs directly into a monitor to output video if your CPU has built-in graphics.
3. **USB 3.0 / 3.2 Ports (Blue, middle):** High-speed USB ports meant for fast external drives, flash drives, or webcams.
4. **Ethernet Port (Top right block):** Where you plug in an RJ-45 internet cable for a wired home network connection.
5. **Audio Jacks (Far right color-coded jacks):** Standard 3.5mm plugs for connecting headphones (green), a microphone (pink), or audio line-in devices (blue).